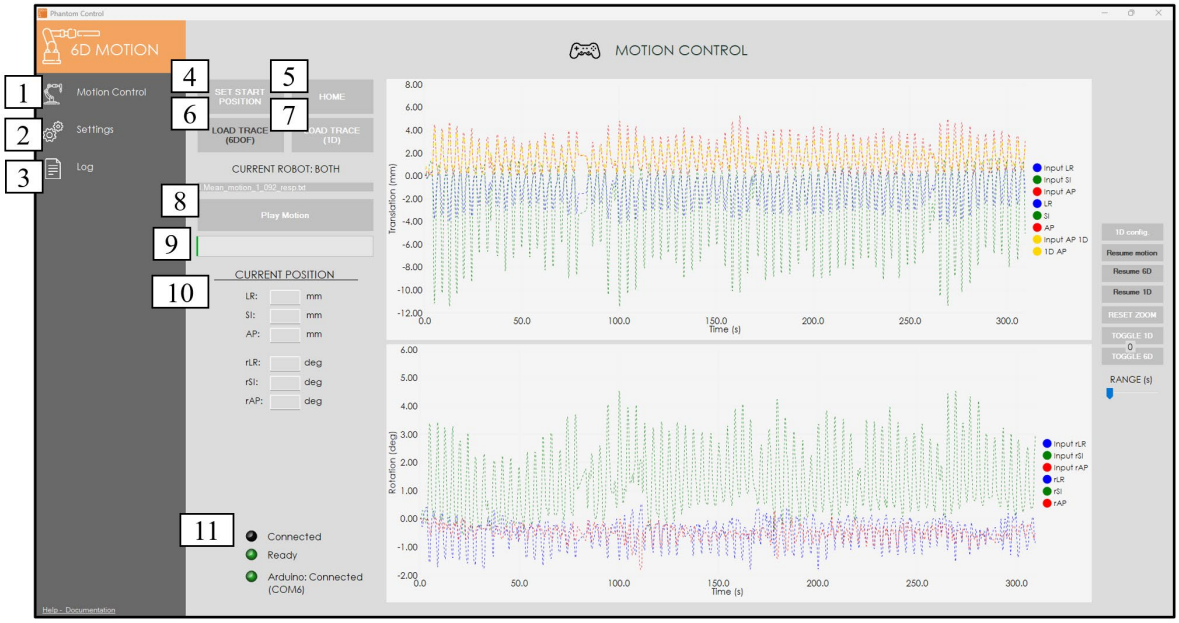
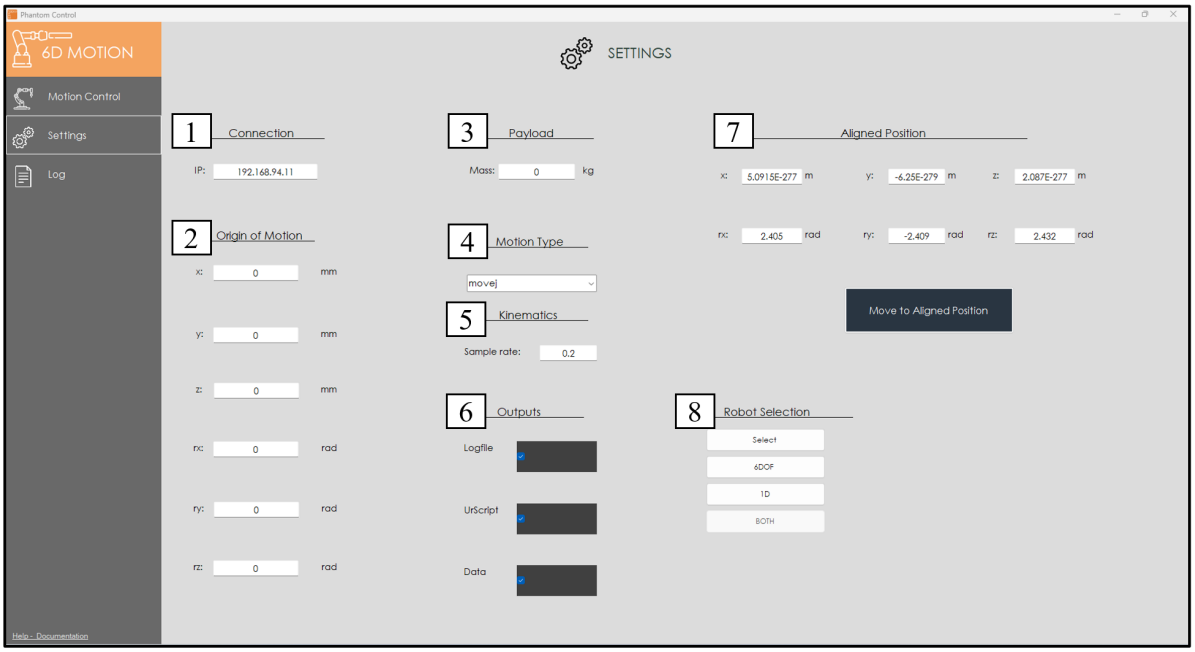
IntERAct QA Device
Software GUI Guide



1. **Motion control tab**
   Opens controls to load traces and run motion (current tab in image above)
2. **Settings tab**
   Opens settings for IntERAct motion platform
3. **Log tab**
   Opens log box to view status of IntERAct motion platform
4. **Set start position**
   Once the 6DoF robotic arm is setup and aligned, click the *Set Start Position* button. This will first establish a connection with the 6DoF robot and get the current (aligned) position of the 6DoF platform.
5. **Home**
   The *Home* button will send the 6DoF motion platform back to the original start position that was set by the *Set Start Position* button. If the 6DoF robot is repositioned and the *Set Start Position* button is clicked again the new *Home* button will be updated to that position.
6. **Load motion trace (6DoF)**
   Once the *Set Start Position* button is clicked and a connection to the 6DoF robot is established, the *Load Motion Trace (6DoF)* button will become active. This will allow a user to load a .txt file in the following format [Time (s) x (mm) y (mm) z (mm) rx (deg) ry (deg) rz (deg)] (space delimited file). The loaded trace will then be displayed on the graph on the right.
7. **Load motion trace (1DoF)**
   Once the *Set Start Position* button is clicked and a connection to the 1DoF motion platform is established, the *Load Motion Trace (1DoF)* button will become active. This will allow a user to load a .txt file in the format [Time (s) z (mm)] (space delimited file). The initial z position needs to be 0 (corresponding to the fully retracted position of the 1DoF platform). The loaded trace will then be displayed on the graph on the right.
8. **Play motion**
   Once the loaded trace(s) is complete and the transformed motion is (automatically) checked for irregular motion (singularity check, large motion check), the *Play Motion* button will become active. Clicking this will send the data to the robot and run the motion. The *Play Motion* button will then change to a *Stop Motion* icon – clicking this will stop the IntERAct motion platform from moving.
9. **Status bar**
   Provides an estimated status of progress of the motion that is running.
10. **Current position value box**

Displays the current position of the 6DoF robot when a motion trace is running.

11. **Display LEDs**
    Provide information if the 6DoF and 1DoF motion platforms are connected and ready for motion. If there is an issue with the motion e.g. a singularity or very large motion, the Ready LED will be red, and the *Play Motion* button will become inactive. If all LEDs are green motion can be run.



1. **Connection**
   Input the IP address of the 6DoF robotic controller for TCP/IP connection.
2. **Origin of motion**
   Sets the origin of motion for both translation and rotation for the input 6DoF robot traces relative to the centre of the end-effector. This is known as the Tool Centre Point (TCP).
3. **Payload**
   Input the weight of the phantom attached to the 6DoF robot in kg including the tool attachment to the end-effector. This is important for the safety of the 6DoF robot and accurate motion.
4. **Motion type**
   Two motion types can be run on the 6DoF robot: movej (default) and movel. Movej makes the robotic arm move from point a to point b in the most efficient way when moving the joints in the arm. Movel makes the robotic arm move from point a to point b linearly, i.e. the joints will move in a way that will force the robot to move in a straight path.
5. **Kinematics**
   The input file sample rate defines the velocity and acceleration of the 6DoF robot. The input sample rate should be the same as the input data sampling frequency (5Hz/0.2 s intervals recommended). This will allow the robot to move from point a to point b in at the set input sampling time.
6. **Outputs**
   Output files can be switched on and off. Logfile outputs a .txt of the logging information during a motion run. UrScript outputs what is sent to the 6DoF robotic controller for executed motion, this is in the robot API language. Data outputs the motion recorded from the 6DoF robot as a .txt file in the format [Time (s) x (mm) y (mm) z (mm) rx (deg) ry (deg) rz (deg)] (space delimited).
7. **Aligned position**
   Once the 6DoF robot (specifically, any attached phantom) is aligned to the treatment isocentre, the user may record the 6DoF position from the tablet of the robot in the Base coordinate system.
8. **Robot selection**
   Setting to select the 6DoF motion platform, 1DoF motion platform, or the combined system ('both')

Settings.txt File Guide

Several settings for the IntERAct system can be changed in the settings text file located in the Release folder. The available settings (as of 16th April 2026) are displayed in Table 1 along with an explanation of each setting.

Table 1. Adjustable settings of the IntERAct system in the settings text file (in the Release folder).

IP Address = 192.168.94.11	IP address of the robot controller.
Phantom Weight [Kg] = 0	Also adjustable in Settings screen of Motion Control Software. 0 kg is the default.
TCPX [mm] = 0	Tool Centre Point offset settings for each DoF of the 6DoF robotic motion platform. The values should reflect the offset from the robot origin to the centre of rotation of the attached tool or phantom. TCP offset settings can also be changed in the Settings screen of the Motion Control Software.
TCPY [mm] = 0	
TCPZ [mm] = 0	
TCPRx [rad] = 0	
TCPRy [rad] = 0	
TCPRz [rad] = 0	
Motion Type = movej	movej (Joint Move) and movel (Linear Move) are primary commands defining how the 6DoF motion platform (robotic arm) reaches a target. MoveJ moves robot joints to target angles, creating a fast, non-linear path ideal for pick-and-place. MoveL drives the tool center point (TCP) in a straight line, used for precise tasks like welding or glueing.
Time [s] = 0.2	Time interval between subsequent time points in the input motion trace files for the 6DoF and 1DoF (or individual) phantoms. Note that if the combined system is used, both platforms need synchronised motion traces at the same frequency.
Output Log File = True	Writes log file for the motion platform. This includes information visible on the Log screen of the Motion Control Software, such as when the mass or TCP offset is changed, or platform is Homed.
Output UrScript File = True	Writes UR Script (robot command information) for the 6DoF robotic motion platform.
Output Data File = True	Writes data file for the motion platform. This includes the timestamp and actuated motion information (robot feedback) for the 6DoF motion platform, and timestamp of start and end of 1DoF platform motion, as well as the motion file and instructions sent to the 1DoF platform.
Aligned PositionX [mm] = 5.0915E-277	A default position of the robot. Clicking “Move to Aligned Position” on the Motion Control Software should theoretically move the robot to this setup position (not if the robot is starting from Home position – use free drive mode to move out of this position first). This functionality has not been recently tested. Please refer to Getting Started documentation for more information: https://github.com/Image-X-
Aligned PositionY [mm] = -6.25E-279	
Aligned PositionZ [mm] = 2.087E-277	
Aligned PositionRx [rad] = 2.405	
Aligned PositionRy [rad] = -2.409	
Aligned PositionRz [rad] = 2.432	

	Institute/IntERAct/blob/main/ Documentation/6DoF/Getting%20Started.pdf
Vertical Volt-speed relationship going up : slope = 0.2089	Calibrated voltage-velocity equation for driving the 1DoF motion platform in the Vertical orientation. Last updated March 2026. Ideally, this should be measured and re-calibrated every 12 months.
Vertical Volt-speed relationship going up : y-intercept = -4.7078	
Vertical Volt-speed relationship going down : slope = 0.2131	
Vertical Volt-speed relationship going down : y-intercept = -6.4044	
Horizontal Volt-speed relationship going up : slope = 0.2800	Calibrated voltage-velocity equation for driving the 1DoF motion platform in the Horizontal orientation. Last updated 2024.
Horizontal Volt-speed relationship going up : y-intercept = -8.2643	
Horizontal Volt-speed relationship going down : slope = 0.2500	
Horizontal Volt-speed relationship going down : y-intercept = -4.3393	
Maximum of 6D input lines = 7300	Setting for the maximum allowable number of input lines in the motion traces. This can be adjusted to allow for higher frequency traces (e.g. 20-50Hz) and needs to be larger than the number of lines in the input trace files. The number of lines can be found in the text input motion trace file or can be calculated as the length of the trace in seconds × input motion frequency (Hz or s^{-1}). E.g. a 300 s long trace with 0.2 s time intervals would need a maximum > $(300 \times (0.2)^{-1})$ = 1500. We recommend a slightly larger number for this setting e.g. 1505. Please note: the motion platform works most efficiently for traces 5 minutes long or less.